

CAP-CHAT, QC, Canada

WMO: 714280

Lat: 49.1092N

Lon: 66.6544W

Elev: 5

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-19.3	-25.8	0.4	-21.5	-23.3	0.5	-18.9	18.7	-8.4	17.2	-9.8	6.9	190	0.776

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	22.8	18.3	21.1	17.4	19.8	16.6	19.2	21.9	18.0	20.6	17.0	19.3	5.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.0	20.8	16.8	12.0	19.6	15.9	11.3	18.7	54.8	22.0	51.1	20.6	48.0	19.4	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
16.0	14.0	12.4	DB	-24.5	27.7	3.1	2.4	-26.7	29.4	-28.5	30.9	-30.3	32.2	-32.5	34.0	
			WB	-25.0	21.6	3.3	1.3	-27.3	22.5	-29.2	23.3	-31.1	24.0	-33.5	24.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-9.7	-9.4	-4.8	1.4	7.3	12.4	16.0	15.5	11.6	6.0	0.8	-5.0
	DBStd	9.75	5.59	5.21	4.88	3.25	3.15	3.10	2.49	2.61	3.21	3.29	4.11	4.66
	HDD10.0	2874	612	543	459	257	98	11	0	1	19	132	276	466
	HDD18.3	5403	871	776	718	507	344	181	81	94	202	382	525	724
	CDD10.0	528	0	0	0	0	13	81	186	171	68	8	2	0
	CDD18.3	16	0	0	0	0	0	2	8	6	1	0	0	0
	CDH23.3	41	0	0	0	0	1	7	16	16	3	0	0	0
	CDH26.7	6	0	0	0	0	0	1	2	3	0	0	0	0
Wind	WSAvg	5.3	6.4	5.9	5.8	5.3	4.3	4.2	4.5	4.7	5.0	5.6	6.0	6.5
Precipitation	PrecAvg	860	64	55	61	55	77	75	82	79	78	95	83	75
	PrecMax	1001	118	99	124	116	139	190	170	170	230	163	191	120
	PrecMin	685	23	27	22	18	34	37	34	16	17	23	32	45
	PrecStd	83	27	21	28	25	31	30	34	40	42	36	39	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	4.3	9.3	13.3	20.4	23.8	25.3	25.1	23.1	19.9	18.0	8.8
		MCWB	5.1	2.8	6.7	7.6	13.7	17.5	19.6	19.6	18.4	15.7	14.4	6.7
	2%	DB	2.0	1.1	4.8	9.2	16.1	20.3	22.7	22.2	20.2	14.6	11.3	4.6
		MCWB	0.7	0.0	2.5	5.6	11.4	15.8	18.5	18.3	16.5	11.7	8.2	3.0
	5%	DB	0.0	-0.4	2.8	6.9	13.7	18.6	21.1	20.6	17.9	12.2	7.5	2.5
		MCWB	-0.7	-1.1	1.1	4.3	10.0	14.7	17.6	17.5	15.0	10.0	5.5	1.4
	10%	DB	-1.7	-1.7	1.2	5.5	11.9	17.0	19.8	19.4	16.1	10.3	5.7	1.0
		MCWB	-2.5	-2.4	-0.1	3.3	8.9	13.7	16.8	16.7	13.8	8.5	4.3	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	3.1	6.9	9.5	15.0	18.9	21.1	20.7	19.6	16.0	14.3	6.9
		MCDB	6.7	4.4	9.0	12.0	19.4	22.7	23.9	23.5	22.5	18.5	18.3	8.5
	2%	WB	1.0	0.3	3.0	5.9	12.2	16.7	19.2	19.0	17.2	11.9	8.8	3.4
		MCDB	1.9	0.9	4.3	8.3	15.0	19.4	21.8	21.9	19.7	14.2	10.9	4.4
	5%	WB	-0.7	-1.0	1.3	4.6	10.6	15.4	18.1	17.9	15.4	10.0	5.9	1.6
		MCDB	-0.1	-0.3	2.6	6.4	13.2	17.9	20.6	20.2	17.5	11.6	7.3	2.4
	10%	WB	-2.6	-2.3	0.2	3.6	9.2	14.2	17.1	16.9	14.1	8.6	4.5	0.4
		MCDB	-1.9	-1.7	1.1	5.2	11.5	16.6	19.4	18.9	16.0	10.0	5.6	1.0
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.0	6.8	5.8	6.6	6.8	6.5	6.2	6.5	5.7	5.4	5.8
		MCDBR	9.7	7.4	7.8	8.6	9.8	9.1	8.4	8.0	9.2	8.7	8.7	8.3
	5% WB	MCWBR	9.6	6.7	6.7	6.4	6.6	6.0	5.6	5.4	6.7	6.8	7.3	7.7
		MCWBR	9.8	7.3	7.3	8.0	9.0	8.4	7.8	7.4	8.8	8.3	8.4	8.0
Clear-Sky Solar Irradiance	taub	taub	0.269	0.278	0.306	0.385	0.365	0.385	0.419	0.388	0.363	0.336	0.296	0.277
		taud	2.360	2.306	2.303	2.218	2.396	2.371	2.313	2.403	2.450	2.500	2.520	2.398
	Ebn at Noon	Ebn at Noon	798	877	907	859	895	875	837	850	838	804	770	737
		Edh at Noon	66	88	106	129	113	117	122	107	92	74	58	57
All-Sky Solar Radiation	RadAvg	1.17	2.11	3.35	4.43	5.20	5.59	5.41	4.65	3.46	2.06	1.07	0.80	
	RadStd	0.14	0.22	0.23	0.39	0.34	0.39	0.55	0.28	0.26	0.17	0.13	0.08	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page